Panskura Banamali College(Autonomous)

M.Sc. 4th Semester Examination -2021 (Online Mode)
Sub: Computer Science Paper: COS-493
Paper Name: Grand Viva
F.M.: 25 Time : 2 Hours

Instruction

[Make your answer script into PDF file format and file name as like candidate_id-paper_code.pdf .
For Example – candidate_id: 2020PG009967, paper_code : COS-493 , then file name will be :: 2020PG009967- COS-493.pdf]

A. Answer any 50 questions (Tick correct option only) : 50*1/2 =25

1. What is an external sorting algorithm?

a) Algorithm that uses tape or disk during the sort
b) Algorithm that uses main memory during the sort
c) Algorithm that involves swapping
d) Algorithm that are considered 'in place'

2. Select the appropriate code that performs bubble sort.

a)

```
for(int j=arr.length-1; j>=0; j--)
{
        for(int k=0; k<j; k++)
        {
                if(arr[k] > arr[k+1])
                {
                        int temp = arr[k];
                        arr[k] = arr[k+1];
                        arr[k+1] = temp;
                }
        }
}
```

b)

```
for(int j=arr.length-1; j>=0; j--)
{
        for(int k=0; k<j; k++)
        {
                if(arr[k] < arr[k+1])
                {
                        int temp = arr[k];
```

```
                        arr[k] = arr[k+1];
                        arr[k+1] = temp;
                }
        }
}
```

c)

```
for(int j=arr.length; j>=0; j--)
{
        for(int k=0; k<j; k++)
        {
                if(arr[k] > arr[k+1])
                {
                        int temp = arr[k];
                        arr[k] = arr[k+1];
                        arr[k+1] = temp;
                }
        }
}
```

d)

```
for(int j=arr.length; j>=0; j--)
{
        for(int k=0; k<j; k++)
        {
                if(arr[k] > arr[k+2])
                {
                        int temp = arr[k];
                        arr[k] = arr[k+1];
                        arr[k+1] = temp;
                }
        }
}
```

3. The given array is arr = {1, 2, 4, 3}. Bubble sort is used to sort the array elements. How many iterations will be done to sort the array?

a) 4

b) 2

c) 1

d) 0

4. How can you improve the best case efficiency in bubble sort? (The input is already sorted)

a)

```
    boolean swapped = false;
        for(int j=arr.length-1; j>=0 && swapped; j--)
        {
                swapped = true;
                for(int k=0; k<j; k++)
                {
                        if(arr[k] > arr[k+1])
```

```
                {
                        int temp = arr[k];
                        arr[k] = arr[k+1];
                        arr[k+1] = temp;
                        swapped = false;
                }
        }
}
```

b)

```
boolean swapped = true;
    for(int j=arr.length-1; j>=0 && swapped; j--)
    {
        swapped = false;
        for(int k=0; k<j; k++)
        {
            if(arr[k] > arr[k+1])
            {
                int temp = arr[k];
                arr[k] = arr[k+1];
                arr[k+1] = temp;
            }
        }
    }
```

c)

```
boolean swapped = true;
    for(int j=arr.length-1; j>=0 && swapped; j--)
    {
        swapped = false;
        for(int k=0; k<j; k++)
        {
            if(arr[k] > arr[k+1])
            {
                int temp = arr[k];
                arr[k] = arr[k+1];
                arr[k+1] = temp;
                swapped = true;
            }
        }
    }
```

d)

```
boolean swapped = true;
    for(int j=arr.length-1; j>=0 && swapped; j--)
    {
        for(int k=0; k<j; k++)
        {
            if(arr[k] > arr[k+1])
            {
                int temp = arr[k];
                arr[k] = arr[k+1];
```

```
                    arr[k+1] = temp;
                    swapped = true;
                }
            }
        }
```

5. The given array is arr = {1,2,4,3}. Bubble sort is used to sort the array elements. How many iterations will be done to sort the array with improvised version?
a) 4
b) 2
c) 1
d) 0

6. If a problem can be solved by combining optimal solutions to non-overlapping problems, the strategy is called _____
a) Dynamic programming
b) Greedy
c) Divide and conquer
d) Recursion

7. When a top-down approach of dynamic programming is applied to a problem, it usually _____
a) Decreases both, the time complexity and the space complexity
b) Decreases the time complexity and increases the space complexity
c) Increases the time complexity and decreases the space complexity
d) Increases both, the time complexity and the space complexity

8. Which of the following problems is NOT solved using dynamic programming?
a) 0/1 knapsack problem
b) Matrix chain multiplication problem
c) Edit distance problem
d) Fractional knapsack problem

9. Which of the following problems should be solved using dynamic programming?
a) Mergesort
b) Binary search
c) Longest common subsequence
d) Quicksort

10. Which of the following is false in the case of a spanning tree of a graph G?
a) It is tree that spans G
b) It is a subgraph of the G
c) It includes every vertex of the G
d) It can be either cyclic or acyclic

11. Consider a complete graph G with 4 vertices. The graph G has ____ spanning trees.
a) 15
b) 8

c) 16
d) 13

12. The travelling salesman problem can be solved using _____
a) A spanning tree
b) A minimum spanning tree
c) Bellman – Ford algorithm
d) DFS traversal

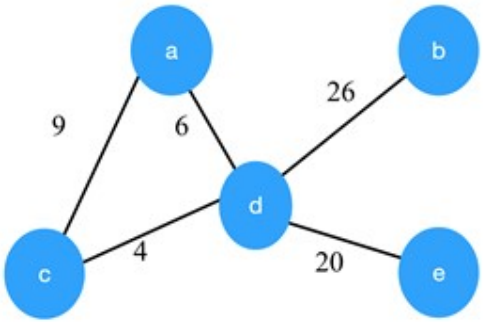
13. Consider the graph M with 3 vertices. Its adjacency matrix is shown below. Which of the following is true?

$$M = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

a) Graph M has no minimum spanning tree
b) Graph M has a unique minimum spanning trees of cost 2
c) Graph M has 3 distinct minimum spanning trees, each of cost 2
d) Graph M has 3 spanning trees of different costs

14. Consider a undirected graph G with vertices { A, B, C, D, E}. In graph G, every edge has distinct weight. Edge CD is edge with minimum weight and edge AB is edge with maximum weight. Then, which of the following is false?
a) Every minimum spanning tree of G must contain CD
b) If AB is in a minimum spanning tree, then its removal must disconnect G
c) No minimum spanning tree contains AB
d) G has a unique minimum spanning tree

15. Consider the graph shown below. Which of the following are the edges in the MST of the given graph?



a) (a-c)(c-d)(d-b)(d-b)
b) (c-a)(a-d)(d-b)(d-e)
c) (a-d)(d-c)(d-b)(d-e)
d) (c-a)(a-d)(d-c)(d-b)(d-e)

16. Which of the following is not the algorithm to find the minimum spanning tree of the given graph?
a) Boruvka's algorithm

b) Prim's algorithm
c) Kruskal's algorithm
d) Bellman–Ford algorithm

17. Which of the following is false?
a) The spanning trees do not have any cycles
b) MST have n – 1 edges if the graph has n edges
c) Edge e belonging to a cut of the graph if has the weight smaller than any other edge in the same cut, then the edge e is present in all the MSTs of the graph
d) Removing one edge from the spanning tree will not make the graph disconnected

18. Dijkstra's Algorithm is used to solve _____ problems.
a) All pair shortest path
b) Single source shortest path
c) Network flow
d) Sorting

19. Which of the following is the most commonly used data structure for implementing Dijkstra's Algorithm?
a) Max priority queue
b) Stack
c) Circular queue
d) Min priority queue

20. What is the time complexity of Dijikstra's algorithm?
a) O(N)
b) $O(N^3)$
c) $O(N^2)$
d) O(logN)

21. Dijkstra's Algorithm cannot be applied on _____
a) Directed and weighted graphs
b) Graphs having negative weight function
c) Unweighted graphs
d) Undirected and unweighted graphs

22. What is the pseudo code to compute the shortest path in Dijkstra's algorithm?
a)

```
if(!T[w].Known)
        if(T[v].Dist + C(v,w) < T[w].Dist)  {
            Decrease(T[w].Dist to T[v].Dist +C(v,w));
            T[w].path=v; }
```

b)

```
if(T[w].Known)
        if(T[v].Dist + C(v,w) < T[w].Dist)  {
            Increase (T[w].Dist to T[v].Dist +C(v,w));
            T[w].path=v; }
```

c)

```
if(!T[w].Known)
        if(T[v].Dist + C(v,w) > T[w].Dist)  {
```

```
            Decrease(T[w].Dist to T[v].Dist +C(v,w);
             T[w].path=v; }
```

d)

```
if(T[w].Known)
        if(T[v].Dist + C(v,w) < T[w].Dist)  {
             Increase(T[w].Dist to T[v].Dist);
            T[w].path=v; }
```

23. How many priority queue operations are involved in Dijkstra's Algorithm?
    a) 1
    b) 3
    c) 2
    d) 4

24. How many times the insert and extract min operations are invoked per vertex?
    a) 1
    b) 2
    c) 3
    d) 0

25. The maximum number of times the decrease key operation performed in Dijkstra's algorithm will be equal to _____
    a) Total number of vertices
    b) Total number of edges
    c) Number of vertices – 1
    d) Number of edges – 1

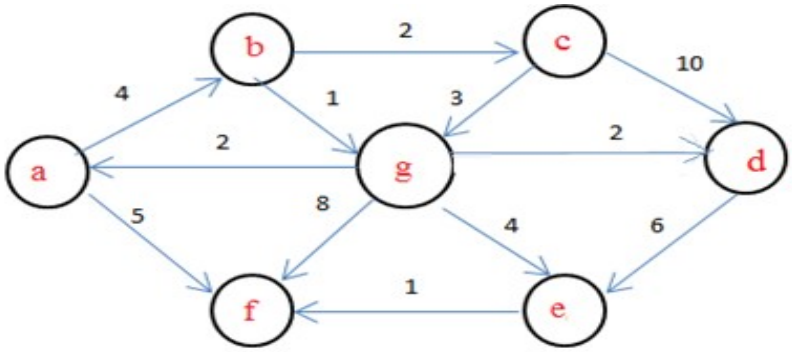
26. What is running time of Dijkstra's algorithm using Binary min- heap method?
    a) O(V)
    b) O(VlogV)
    c) O(E)
    d) O(ElogV)

27. Given pseudo code of Dijkstra's Algorithm.

```
//Initialise single source(G,s)
S=0
Q=V[G]
While Q != 0
        Do u=extract-min(Q)
                S=S union {u}
                For each vertex v in adj[u]
                        Do relax(u,v,w)
```

What happens when while loop in line 4 is changed to while Q>1?
a) While loop gets executed for v times
b) While loop gets executed for v-1 times
c) While loop gets executed only once
d) While loop does not get executed

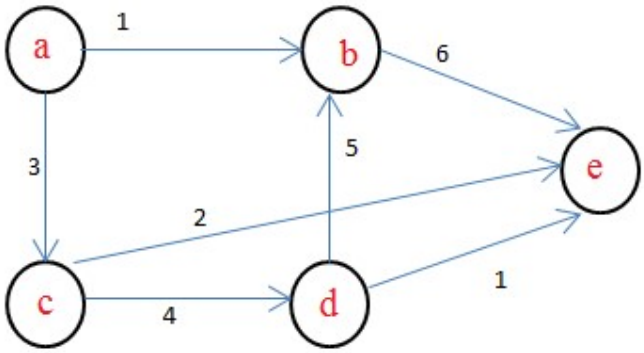
28. Consider the following graph.



If b is the source vertex, what is the minimum cost to reach f vertex?

a) 8
b) 9
c) 4
d) 6

29. In the given graph:



Identify the shortest path having minimum cost to reach vertex E if A is the source vertex

a) a-b-e
b) a-c-e
c) a-c-d-e
d) a-c-d-b-e

30. Dijkstra's Algorithm is the prime example for _____

a) Greedy algorithm
b) Branch and bound
c) Back tracking
d) Dynamic programming

31. _____ has the lowest fault rate of all the page replacement algorithms.

a) Optimal page replacement algorithm
b) LRU replacement algorithm
c) FIFO
d) Counting based

32. Optimal page replacement algorithm is also called as _____

a) LIFO

b) NRU
c) Clairvoyant replacement algorithm
d) Page buffering

33. In a optimal page replacement algorithm, when a page is to be replaced, which of the following pages is chosen?
a) Oldest page
b) Newest page
c) Frequently occurred page in the future
d) Not frequently occurred page in the future

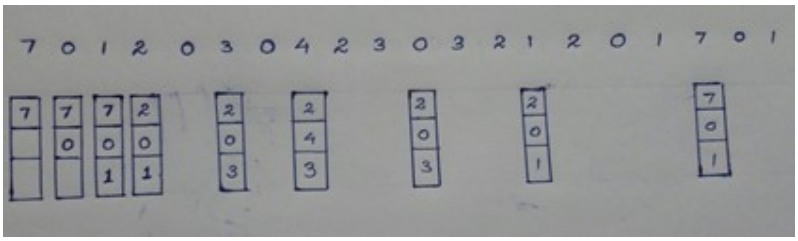
34. Analysis of the optimal paging problem has been done through_____
a) Deterministic algorithm
b) Online algorithm
c) Euclid algorithm
d) Optimal algorithm

35. Optimal page replacement algorithm is implemented in _____
a) General-purpose operating system
b) Special-purpose operating system
c) In any kind of operating system
d) In Windows only

36. In a stack algorithm, the set of pages in a k-frame memory is always a subset of pages in a _____ frame memory.
a) k-1
b) k
c) k+1
d) k(k+1)

37. Consider a reference string 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1 of frame size 3. Calculate the number of page faults using optimal page replacement algorithm.
a) 10
b) 9
c) 8
d) 7

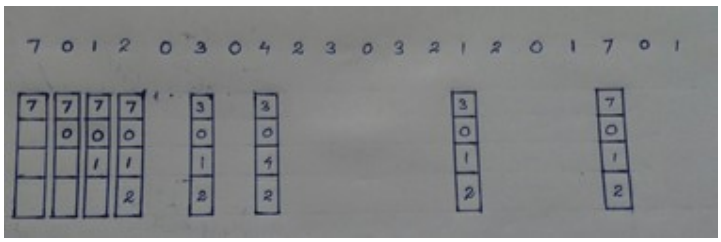
38. Consider a reference string 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1 of frame size 4. Calculate the number of page faults using optimal page replacement algorithm.



a) 7
b) 9

c) 8
d) 6

39. Consider a reference string 1,2,3,2,1,5,2,1,6,2,5,6,3,1,3,6,1,2,4,3 of frame size 3. Calculate the number of page faults using optimal page replacement algorithm.



a) 12
b) 16
c) 14
d) 15

40. Recursion is a method in which the solution of a problem depends on _____
a) Larger instances of different problems
b) Larger instances of the same problem
c) Smaller instances of the same problem
d) Smaller instances of different problems

41. Consider the following code snippet:

```
void my_recursive_function()
{
  my_recursive_function();
}
int main()
{
  my_recursive_function();
  return 0;
}
```

What will happen when the above snippet is executed?
a) The code will be executed successfully and no output will be generated
b) The code will be executed successfully and random output will be generated
c) The code will show a compile time error
d) The code will run for some time and stop when the stack overflows

42. What is the output of the following code?

```
void my_recursive_function(int n)
{
   if(n == 0)
   return;
   printf("%d ",n);
   my_recursive_function(n-1);
}
int main()
```

```
{
   my_recursive_function(10);
   return 0;
}
```

a) 10
b) 1
c) 10 9 8 … 1 0
d) 10 9 8 … 1

43. What is the base case for the following code?

```
void my_recursive_function(int n)
{
    if(n == 0)
    return;
    printf("%d ",n);
    my_recursive_function(n-1);
}
int main()
{
    my_recursive_function(10);
    return 0;
}
```

a) return
b) printf(“%d “, n)
c) if(n == 0)
d) my_recursive_function(n-1)

44. How many times is the recursive function called, when the following code is executed?

```
void my_recursive_function(int n)
{
    if(n == 0)
    return;
    printf("%d ",n);
    my_recursive_function(n-1);
}
int main()
{
    my_recursive_function(10);
    return 0;
}
```

a) 9
b) 10
c) 11
d) 12

45. What does the following recursive code do?

```
void my_recursive_function(int n)
{
    if(n == 0)
    return;
    my_recursive_function(n-1);
    printf("%d ",n);
}
int main()
```

```
{
    my_recursive_function(10);
    return 0;
}
```

a) Prints the numbers from 10 to 1
b) Prints the numbers from 10 to 0
c) Prints the numbers from 1 to 10
d) Prints the numbers from 0 to 10

46. Which of the following statements is true?
a) Recursion is always better than iteration
b) Recursion uses more memory compared to iteration
c) Recursion uses less memory compared to iteration
d) Iteration is always better and simpler than recursion

47. What will be the output of the following code?

```
int cnt=0;
void my_recursive_function(int n)
{
    if(n == 0)
    return;
    cnt++;
    my_recursive_function(n/10);
}
int main()
{
    my_recursive_function(123456789);
    printf("%d",cnt);
    return 0;
}
```

a) 123456789
b) 10
c) 0
d) 9

48. What will be the output of the following code?

```
void my_recursive_function(int n)
{
   if(n == 0)
   {
       printf("False");
           return;
   }
   if(n == 1)
   {
       printf("True");
       return;
```

```
    }
    if(n%2==0)
    my_recursive_function(n/2);
    else
    {
        printf("False");
        return;
    }

}
int main()
{
    my_recursive_function(100);
    return 0;
}
```

a) True
b) False
c) Both
d) None

49. What is the output of the following code?

```
int cnt = 0;
void my_recursive_function(char *s, int i)
{
    if(s[i] == '\0')
      return;
    if(s[i] == 'a' || s[i] == 'e' || s[i] == 'i' || s[i] == 'o' || s[i] == 'u')
    cnt++;
    my_recursive_function(s,i+1);
}
int main()
{
    my_recursive_function("thisisrecursion",0);
    printf("%d",cnt);
    return 0;
}
```

a) 6
b) 9
c) 5
d) 10

50. What is the output of the following code?

```
void my_recursive_function(int *arr, int val, int idx, int len)
{
    if(idx == len)
    {
        printf("-1");
        return ;
    }
    if(arr[idx] == val)
```

```
    {
        printf("%d",idx);
        return;
    }
   my_recursive_function(arr,val,idx+1,len);
}
int main()
{
    int array[10] = {7, 6, 4, 3, 2, 1, 9, 5, 0, 8};
    int value = 2;
    int len = 10;
    my_recursive_function(array, value, 0, len);
    return 0;
}
```

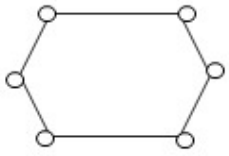
a) 3
b) 4
c) 5
d) 6

51. The Data structure used in standard implementation of Breadth First Search is?
    a) Stack
    b) Queue
    c) Linked List
    d) Tree

52. When the Depth First Search of a graph is unique?
    a) When the graph is a Binary Tree
    b) When the graph is a Linked List
    c) When the graph is a n-ary Tree
    d) When the graph is a ternary Tree

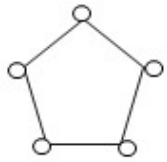
53. When the Breadth First Search of a graph is unique?
    a) When the graph is a Binary Tree
    b) When the graph is a Linked List
    c) When the graph is a n-ary Tree
    d) When the graph is a Ternary Tree

54. What will be the chromatic index of the following graph?



    a) 1
    b) 2
    c) 3
    d) 4

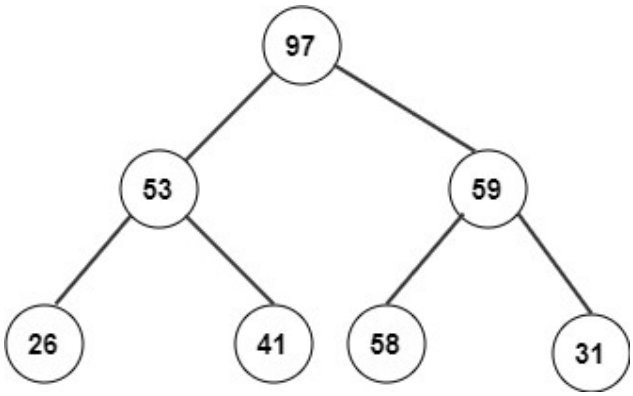
55. What will be the chromatic index of the following graph?



a) 2
b) 3
c) 4
d) 5

56. What will be the chromatic index for a complete graph having n vertices (consider n to be an even number)?
a) n
b) n + 1
c) n – 1
d) 2n + 1

57. Consider the following heap after build heap phase. What will be its corresponding array?



a) 26,53,41,97,58,59,31
b) 26,31,41,53,58,59,97
c) 26,41,53,97,31,58,59
d) 97,53,59,26,41,58,31

58. What is the average number of comparisons used to heap sort a random permutation of N distinct items?
a) 2N log N-O(N)
b) 2N log N-O(N log N)
c) 2N log N-O(N log log N)
d) 2N log N-O(log N)

59. Consider the original array 17 8 12 4 26. How many comparisons are needed to construct the BST on the original array?
a) 5
b) 4

c) 7
d) 10

60. In binary tree sort, we first construct the BST and then we perform _____ traversal to get the sorted order.
a) inorder
b) postorder
c) preorder
d) level order

61. The insert() procedure, given below, builds the BST on the input elements, which is the first step of the binary tree sort. Choose the correct to fill the condition.

```
void insert(Tree* node, int newElement)
{
        if(node== NULL)
        {
                node = createNewNode();
                node-> value = newElement;
                node -> left = NULL;
                node -> right = NULL;
                return;
        }
        else if(_____)
        {
                insert(node->left, newElement);
        }
        else
        {
                insert(node->right, newElement);
        }
}
```

a) newElement > node->value
b) newElement < node->value
c) newElement == root->value
d) newElement != root->value

62. Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?
a) 90 and 99
b) 90 and 94
c) 89 and 99
d) 89 and 94

63. Given an array arr = {5,6,77,88,99} and key = 88; How many iterations are done until the element is found?
a) 1
b) 3
c) 4
d) 2

64. Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?
a) 90 and 99
b) 90 and 100
c) 89 and 94
d) 94 and 99

65. If the value in protocol field is 17, the transport layer protocol used is _____
a) TCP
b) UDP
c) ICMP
d) IGMP

66. Which of these is not applicable for IP protocol?
a) is connectionless
b) offer reliable service
c) offer unreliable service
d) does not offer error reporting

67. Which NetWare protocol works on layer 3–network layer—of the OSI model?
a) IPX
b) NCP
c) SPX
d) NetBIOS

68. Which of the following is not correct in relation to multi-destination routing?
a) is same as broadcast routing
b) contains the list of all destinations
c) data is not sent by packets
d) there are multiple receivers

69. In TCP IP Model, when data is sent from device A to device B, the 5th layer to receive data at B is _____
a) Application layer
b) Transport layer
c) Link layer
d) Session layer

70. A 4 byte IP address consists of _____
a) only network address
b) only host address
c) network address & host address
d) network address & MAC address

71. The network layer protocol for internet is _____
a) ethernet
b) internet protocol
c) hypertext transfer protocol
d) file transfer protocol

72. In an IPv6 header, the traffic class field is similar to which field in the IPv4 header?
a) Fragmentation field
b) Fast switching
c) TOS field
d) Option field

73. IPv6 does not use _____ type of address.
a) Broadcast
b) Multicast
c) Any cast
d) Unicast

74. Which are the features present in IPv4 but not in IPv6?
a) Fragmentation
b) Header checksum
c) Options
d) Anycast address

75. Suppose a TCP connection is transferring a file of 1000 bytes. The first byte is numbered 10001. What is the sequence number of the segment if all data is sent in only one segment?
a) 10000
b) 10001
c) 12001
d) 11001

76. Bytes of data being transferred in each connection are numbered by TCP. These numbers start with a _____
a) Fixed number
b) Random sequence of 0's and 1's
c) One
d) Sequence of zero's and one's

77. What is the stored in the object obj in following lines of Java code?

```
box obj;
```

a) Memory address of allocated memory of object
b) NULL
c) Any arbitrary pointer
d) Garbage

What will be the output of the following Java program?

```
class main_class
{
   public static void main(String args[])
   {
      int x = 9;
      if (x == 9)
      {
         int x = 8;
         System.out.println(x);
      }
   }
}
```

a) 9
b) 8
c) Compilation error
d) Runtime error

78. What will be the output of the following Java program?

```
class box
{
   int width;
   int height;
   int length;
}
class mainclass
{
   public static void main(String args[])
   {
      box obj1 = new box();
      box obj2 = new box();
      obj1.height = 1;
      obj1.length = 2;
      obj1.width = 1;
      obj2 = obj1;
      System.out.println(obj2.height);
   }
}
```

a) 1
b) 2
c) Runtime error
d) Garbage value

79. 3. The 10's complement of 562 is _____

a) 4
b) 3

c) 7
d) 8

80. The result obtained on subtraction using 2's complement of 1111-0010 will be _____
a) 11101
b) 11011
c) 1101
d) 1011

81. Object=_____+relationships.
a) data
b) attributes
c) entity
d) constraints
Answer: c
Explanation: The answer is entity. It is a part of OODBM (Object-Oriented Database Model). It maintains the advantages of ER-model but adds more features.

82. Database is generally _____
a) System-centered
b) User-centered
c) Company-centered
d) Data-centered

83. SQL is _____
a) Relational
b) Network
c) IMS
d) Hierarchical

84. DBMS is a set of _____ to access the data.
a) Codes
b) Programs
c) Information
d) Metadata

85. Which of the following statements is false?
a) Left as well as right most derivations can be in Unambiguous grammar
b) An LL (1) parser is a top-down parser
c) LALR is more powerful than SLR
d) Ambiguous grammar can't be LR (k)

86. A group of bits that tell the computer to perform a specific operation is known as _____
a) Instruction code
b) Micro-operation

c) Accumulator
d) Register

87. What is the similarity between LR, LALR and SLR?
a) Use same algorithm, but different parsing table
b) Same parsing table, but different algorithm
c) Their Parsing tables and algorithm are similar but uses top down approach
d) Both Parsing tables and algorithm are different

88. The non- Kleene Star operation accepts the following string of finite length over set A = {0,1} | where string s contains even number of 0 and 1
a) 01,0011,010101
b) 0011,11001100
c) ε,0011,11001100
d) ε,0011,11001100

89. A regular language over an alphabet ∑ is one that cannot be obtained from the basic languages using the operation
a) Union
b) Concatenation
c) Kleene*
d) All of the mentioned

90. Given: ∑= {a, b}
L= {xϵ∑*|x is a string combination}
∑4 represents which among the following?

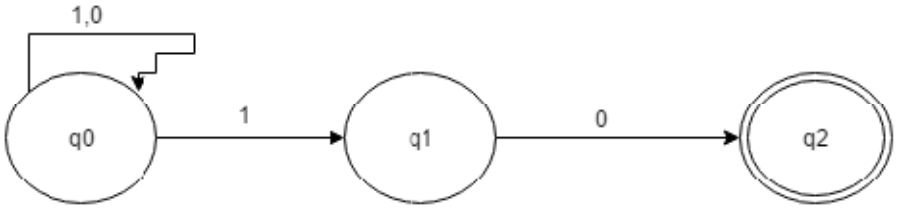
a) {aa, ab, ba, bb}
b) {aaaa, abab, ε, abaa, aabb}
c) {aaa, aab, aba, bbb}
d) All of the mentioned

91. A->aA| a| b
The number of steps to form aab:
a) 2
b) 3
c) 4
d) 5

92. Which among the following is the correct option for the given grammar?
G->X111|G1,X->X0|00
a) $\{0^a1^b|a=2,b=3\}$
b) $\{0^a1^b|a=1,b=5\}$
c) $\{0^a1^b|a=b\}$
d) More than one of the mentioned is correct

93. Which of the following the given language belongs to?
$L=\{a^m b^m c^m | m>=1\}$
a) Context free language
b) Regular language
c) Both (a) and (b)
d) None of the mentioned

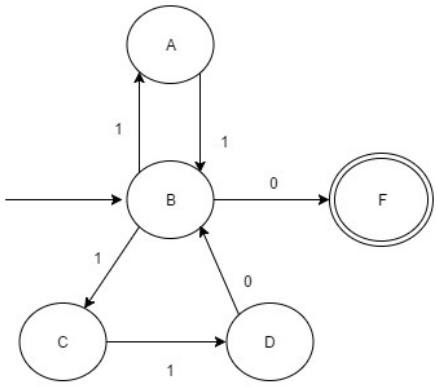
94. The number of transitions required to convert the following into equivalents DFA:



a) 2
b) 3
c) 1
d) 0

95. Given Language:
$L_n = \{x \epsilon \{0,1\} * | |x| \geq n$, nth symbol from the right in x is 1$\}$
How many state are required to execute L_3 using NFA?
a) 16
b) 15
c) 8
d) 7

96. Which of the following does the given NFA represent?



a) {11, 101} * {01}
b) {110, 01} * {11}
c) {11, 110} * {0}
d) {00, 110} * {1}

97. _____ Is an algorithm, a loop that continually moves in the direction of increasing value – that is uphill.
a) Up-Hill Search

b) Hill-Climbing
c) Hill algorithm
d) Reverse-Down-Hill search

98. The correct sequence of GCC compilation process is
a) preprocessing -> compilation -> assemble -> linking
b) assemble -> preprocessing -> compilation -> linking
c) preprocessing -> assemble -> compilation -> linking
d) none of the mentioned

99. If we do not specify the executable file name at the compilation time in GCC, then in linux the compiler creates executable named as
a) a.out
b) a.exe
c) x.out
d) x.exe

100. The expected value for the estimation variable (size), S, can be computed as a weighted average of the optimistic(Sopt), most likely (Sm), and pessimistic (Spess) estimates given as
a) EV = (Sopt + 4Sm + Spess)/4
b) EV = (Sopt + 4Sm + Spess)/6
c) EV = (Sopt + 2Sm + Spess)/6
d) EV = (Sopt + 2Sm + Spess)/4

***